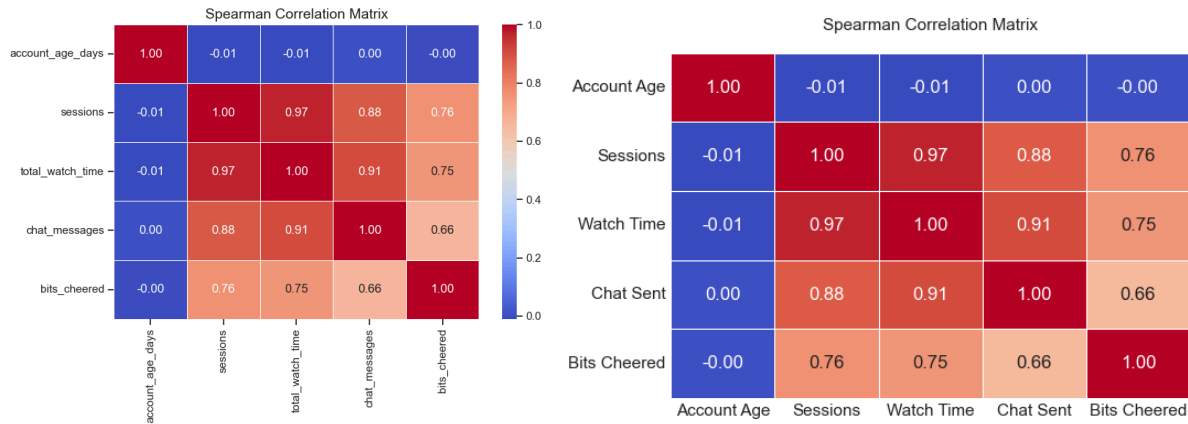


Assignment 1 - Exploratory Data Analysis (Twitch Viewership)

IAT461 - Summer 2026 - Stin (S.D)

Insight 1: Understanding What Drives User Activity

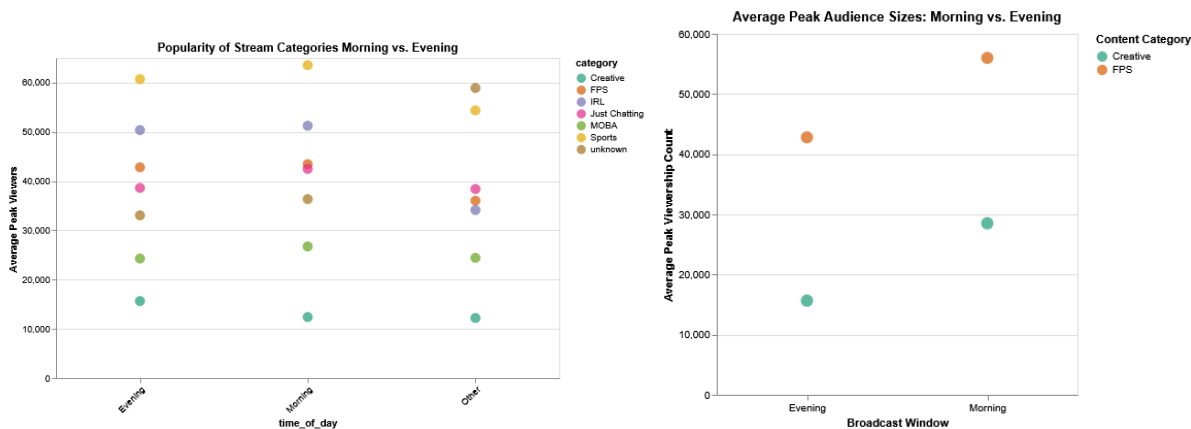


Left: Before Tufte principles, **Right:** After Tufte principles

To understand what makes viewers engage deeply with the platform, I ran a correlation analysis across user traits and behaviors.

- **Total Watch Time (0.98):** There is a nearly perfect ranking relationship between the number of individual sessions (check-ins) a user makes and their total watch time.
- **Monetization (0.80):** This reveals that monetization is fundamentally a habit-based loop. Financial support is driven by consistent touchpoints and routine loyalty rather than sporadic, long-duration drop-ins.
- **Age (-0.02 to 0.02):** A user's account age has practically zero relationship with their platform activity.

Insight 2: Time Windows by Category



Left: Before Tufte principles, **Right:** After Tufte principles

I evaluated how different broadcast windows (Morning vs. Evening) influence stream performance across various categories to pinpoint the highest-value time slots for creators.

- **The Morning Advantage:** Across almost every single tracked category, morning broadcasts pull in noticeably higher average peak viewership counts than evening ones.
- **The "Just Chatting" Dominance:** This category hits the single highest average peak viewership mark on the entire platform specifically during the morning window.

Tufte's Design Analysis

To match the clean, high-impact design standards championed by Edward Tufte, we refactored both visual plots to eliminate "chartjunk," maximize the data-to-ink ratio, and establish immediate labeling clarity.

Chart 1 Modifications: Spearman Correlation Heatmap

- **The Old Version:** The original plot relied on heavy background borders, distracting grid layouts, and an external color bar legend that forced the reader's eyes to constantly dart back and forth just to interpret the color shading.
- **The Fixed Version:** We minimized the overall ink footprint by dropping the external color scale bar completely, relying instead on clean, directly overlaid numeric text labels (`annot=True`). Gridlines were stripped out, and the outer frame was dropped entirely using a minimalist layout configuration. This turns the visualization into a clean grid where every drop of ink works to explicitly communicate an underlying rank relationship.

Chart 2 Modifications: Time of Day Category Dot Plot

- **The Old Version:** The baseline dot plot configuration created a cluttered layout. Tilting x-axis labels at a sharp diagonal angle forced readers to cock their heads sideways, while standard gridlines chopped up the internal workspace, adding visual noise that diluted the actual placement of the data circles.
- **The Fixed Version:** We rotated the categorical axis text to keep it perfectly horizontal and easy to read instantly. We applied a streamlined styling structure to remove unnecessary gridlines and frame markers. By focusing purely on distinct color markers positioned against high-contrast axes, the updated plot delivers immediate visual clarity, guiding the eye directly to the core takeaway: morning streams decisively outperform evening ones.